



VanTech - Data Solutions

Sistema de Monitoramento de Espelho de Ponto

Código Fonte Completo — Documentação Técnica

Tecnologia: HTML5 · CSS3 · JavaScript (Vanilla) · Chart.js 4.4.1

Arquitetura: Single Page Application (SPA) — arquivo único auto-contido

Colaboradores: 5.030 registros gerados proceduralmente (seed determinístico)

Competências: Maio/2026 e Junho/2026

Gerado em: 29/06/2026 às 22:33

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2.	Estrutura HTML — Layout e Páginas
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1. Visão Geral da Arquitetura

O sistema foi desenvolvido como uma **Single Page Application (SPA)** contida em um único arquivo HTML auto-suficiente. Não há dependências de backend, banco de dados ou build tools — toda a lógica, dados e estilos estão encapsulados no arquivo **vantech-ponto.html**. A única dependência externa é a biblioteca Chart.js, carregada via CDN.

■ Toda a massa de dados (5.030 colaboradores, histórico de ponto, ocorrências) é gerada em tempo de execução por um gerador pseudo-aleatório determinístico baseado em seed. Os mesmos dados são reproduzidos identicamente a cada carregamento da página.

Estrutura Geral do Arquivo

<style>	Design System completo: variáveis CSS, layout sidebar/main, tabelas, cards, modal, estilos do espelho de ponto
HTML body	Sidebar fixa com navegação, topbar, 6 páginas (.page) com display:none/block, modal do espelho
<script>	Geração de dados, lógica de ponto, renderização de tabelas, gráficos Chart.js, exportação PDF/CSV

Páginas do Sistema

Dashboard	Cards KPI + 7 gráficos Chart.js com dados do período selecionado
Colaboradores	Tabela paginada (25/página) com busca, 4 filtros, ordenação e botão "Ver espelho"
Pendências	KPIs de assinatura + tabela de espelhos aguardando ação
Exportar Espelhos	Painel de busca por matrícula/nome/setor, seleção múltipla, exportação PDF/CSV em lote
Jornadas	Tabela de jornadas cadastradas + cards de regras de horas extras e banco de horas
Parâmetros	Configurações da empresa, apuração, assinatura digital e exportação

2. Estrutura HTML — Layout e Páginas

O layout usa **Flexbox** com sidebar fixa de 240px e área main com margin-left correspondente. Cada página é um `<div class="page">` que recebe a classe `active` via JavaScript para exibição (`display:block`), ocultando as demais (`display:none`).

Sidebar — Navegação

```
<!-- SIDEBAR -->
<nav id="sidebar">
  <div class="sidebar-logo">
  <div class="sidebar-section">
    <div class="sidebar-label">Principal</div>
    <div class="sidebar-item active" onclick="navigate('dashboard')">
    <div class="sidebar-item" onclick="navigate('colaboradores')">
    <span class="sidebar-badge">5.030</span>
    <div class="sidebar-item" onclick="navigate('pendencias')">
    <span class="sidebar-badge">347</span>
    <div class="sidebar-section">
    <div class="sidebar-label">Relatórios</div>
    <div class="sidebar-item" onclick="navigate('exportar')">
    <div class="sidebar-section">
    <div class="sidebar-label">Configuração</div>
    <div class="sidebar-item" onclick="navigate('jornadas')">
    <div class="sidebar-item" onclick="navigate('parametros')">
    <div class="sidebar-footer">
```

Topbar — Seletor de Competência

O seletor de competência (Maio/2026 | Junho/2026) fica na topbar e chama **setComp(c)** ao clicar, alternando a classe **active** entre os botões e regenerando todos os dados da tela atual.

Modal do Espelho de Ponto

O modal é um `<div id="mirror-overlay">` com `position:fixed` e `display:none`. Ao clicar em "Ver" em qualquer colaborador, a função **openMirror(empIdx)** constrói o HTML completo do espelho via **buildMirrorHTML(emp)** e injeta no `#modal-body`.

3. CSS — Design System

O CSS utiliza **variáveis customizadas (CSS Custom Properties)** no :root para manter consistência de cores, sombras e bordas em todo o sistema. O design segue padrões de sistemas corporativos como TOTVS RM e SAP SuccessFactors.

Variáveis do Design System

```
:root{--brand:#1a56db;--brand-dark:#1245b8;--sidebar-bg:#0f172a;--sidebar-text:#94a3b8;--surface:#fff;--surface-2:#f8fafc;--border:#e2e8f0;--text:#0f172a;--text-2:#64748b;--success:#10b981;--warning:#f59e0b;--danger:#ef4444;--radius:8px;--shadow:0 1px 3px rgba(0,0,0,.08),0 1px 2px rgba(0,0,0,.06);--shadow-lg:0 10px 15px rgba(0,0,0,.08),0 4px 6px rgba(0,0,0,.05)}

body{font-family:'Segoe UI',system-ui,sans-serif;background:#f1f5f9;color:var(--text);display:flex;min-height:100vh;font-size:13px}

#sidebar{width:240px;min-height:100vh;background:var(--sidebar-bg);display:flex;flex-direction:column;position:fixed;left:0;top:0;bottom:0;z-index:100}

.logo-icon{width:32px;height:32px;background:var(--brand);border-radius:8px;display:flex;align-items:center;justify-content:center;font-weight:800;color:#fff;font-size:14px}

.logo-sub{color:var(--sidebar-text);font-size:10px}

.sidebar-item{display:flex;align-items:center;gap:10px;padding:9px 16px;color:var(--sidebar-text);cursor:pointer;border-left:3px solid transparent;transition:all .15s;font-size:13px}

.sidebar-item.active{color:#fff;background:rgba(26,86,219,.15);border-left-color:var(--brand)}

.sidebar-badge{margin-left:auto;background:var(--brand);color:#fff;font-size:9px;padding:2px 6px;border-radius:10px;font-weight:700}

.user-avatar{width:30px;height:30px;background:var(--brand);border-radius:50%;display:flex;align-items:center;justify-content:center;color:#fff;font-size:11px;font-weight:700;flex-shrink:0}

.topbar{background:#fff;border-bottom:1px solid var(--border);padding:0 24px;height:56px;display:flex;align-items:center;gap:16px;position:sticky;top:0;z-index:50}

.topbar-sep{color:var(--border)}

.topbar-sub{color:var(--text-2);font-size:13px}

.comp-selector{display:flex;align-items:center;background:var(--surface-2);border:1px solid var(--border);border-radius:var(--radius);overflow:hidden}

.comp-btn{padding:6px 14px;font-size:12px;font-weight:500;cursor:pointer;color:var(--text-2);background:none;border:none;transition:all .15s}

.comp-btn.active{background:var(--brand);color:#fff}

.icon-btn{width:34px;height:34px;border:1px solid var(--border);border-radius:var(--radius);background:none;cursor:pointer;display:flex;align-items:center;justify-content:center;color:var(--text-2)}

.page-desc{color:var(--text-2);font-size:13px;margin-top:2px}

.card{background:var(--surface);border:1px solid var(--border);border-radius:var(--radius);padding:16px;box-shadow:var(--shadow)}

.card-label{font-size:11px;font-weight:600;color:var(--text-2);text-transform:uppercase;letter-spacing:.5px}

.card-delta{font-size:11px;margin-top:6px;color:var(--text-2)}

.delta-up{color:var(--success)}.color-danger{color:var(--danger)}.color-warning{color:var(--warning)}.color-success{color:var(--success)}

.chart-card{background:var(--surface);border:1px solid var(--border);border-radius:var(--radius);padding:16px;box-shadow:var(--shadow)}

.chart-sub{font-size:11px;color:var(--text-2);margin-bottom:14px}

.table-card{background:var(--surface);border:1px solid var(--border);border-radius:var(--radius);box-shadow:var(--shadow);overflow:hidden}

.table-toolbar{padding:14px 16px;display:flex;align-items:center;gap:10px;border-bottom:1px solid var(--border);flex-wrap:wrap}

.search-box{display:flex;align-items:center;gap:8px;border:1px solid var(--border);border-radius:var(--radius);padding:7px 12px;background:var(--surface-2);flex:1;max-width:320px}

.search-box input{border:none;background:none;outline:none;font-size:13px;color:var(--text);width:100%}

.search-box svg{width:15px;height:15px;color:var(--text-2);flex-shrink:0}
```

```

.filter-select{border:1px solid var(--border);border-radius:var(--radius);padding:7px
10px;font-size:12px;background:var(--surface);color:var(--text);outline:none;cursor:pointer}
.btn{display:inline-flex;align-items:center;gap:6px;padding:7px
14px;border-radius:var(--radius);font-size:12px;font-weight:500;cursor:pointer;border:none;transition:all
.15s,white-space:nowrap}
.btn-primary{background:var(--brand);color:#fff}.btn-primary:hover{background:var(--brand-dark)}
.btn-secondary{background:var(--surface-2);border:1px solid
var(--border);color:var(--text)}.btn-secondary:hover{background:#e2e8f0}
th{background:var(--surface-2);font-size:11px;font-weight:700;color:var(--text-2);text-transform:uppercase;lett
er-spacing:.5px;padding:10px 14px;text-align:left;border-bottom:1px solid
var(--border);white-space:nowrap;cursor:pointer;user-select:none}
td{padding:10px 14px;border-bottom:1px solid var(--border);font-size:12px;color:var(--text);white-space:nowrap}
.mat-code{font-family:monospace;font-size:11px;color:var(--brand);font-weight:600}
.pagination{display:flex;align-items:center;justify-content:space-between;padding:12px 16px;border-top:1px
solid var(--border)}
.pagination-info{font-size:12px;color:var(--text-2)}
.page-btn{width:30px;height:30px;border:1px solid var(--border);border-radius:6px;background:none;cursor:pointe
r;display:flex;align-items:center;justify-content:center;font-size:12px;color:var(--text-2);transition:all
.15s}
.page-btn.active{background:var(--brand);color:#fff;border-color:var(--brand)}
.page-btn:hover:not(.active):not(:disabled){background:var(--surface-2)}
.info-strip{background:#eff6ff;border:1px solid #bfbdbf;border-radius:var(--radius);padding:10px
14px;font-size:12px;color:#1e40af;display:flex;align-items:center;gap:8px;margin-bottom:16px}
.kpi{background:#fff;border:1px solid var(--border);border-radius:var(--radius);padding:12px
16px;flex:1;min-width:140px}
.kpi-label{font-size:10px;font-weight:700;color:var(--text-2);text-transform:uppercase;letter-spacing:.5px}
.kpi-sub{font-size:10px;color:var(--text-2)}
#mirror-modal{background:#fff;border-radius:10px;width:100%;max-width:980px;max-height:92vh;display:flex;flex-d
irection:column;box-shadow:var(--shadow-lg)}
.modal-header{padding:16px 20px;border-bottom:1px solid
var(--border);display:flex;align-items:center;justify-content:space-between;flex-wrap:wrap;gap:8px}
.export-panel{background:#fff;border:1px solid
var(--border);border-radius:var(--radius);box-shadow:var(--shadow);overflow:hidden;margin-bottom:20px}
.export-search-desc{font-size:12px;color:var(--text-2);margin-bottom:16px}
.export-label{font-size:10px;font-weight:700;color:var(--text-2);text-transform:uppercase;letter-spacing:.5px}
.export-results-bar{padding:10px 16px;background:#fafbfc;border-top:1px solid var(--border);border-bottom:1px
solid var(--border);display:flex;align-items:center;justify-content:space-between;flex-wrap:wrap;gap:8px}
.param-card{background:#fff;border:1px solid
var(--border);border-radius:var(--radius);padding:16px;box-shadow:var(--shadow)}
.param-card-title{font-size:12px;font-weight:700;color:var(--text-2);text-transform:uppercase;letter-spacing:.5
px;margin-bottom:12px;padding-bottom:8px;border-bottom:1px solid var(--border)}
.param-row{display:flex;justify-content:space-between;align-items:center;padding:7px 0;border-bottom:1px solid
var(--border);font-size:12px}
.param-key{color:var(--text-2)}
.param-val{font-weight:600;color:var(--text)}

```

Classes Principais

card / cards-grid	Cards KPI em grid de 4 colunas com ícones coloridos e valores numéricos
badge / badge-*	Pílulas de status: verde=ativo, vermelho=desligado, amarelo=férias
table-card	Container de tabela com toolbar, scroll horizontal, paginação integrada
ep-wrap / ep-table	Espelho de ponto fiel ao modelo corporativo — layout idêntico ao insoft4
sidebar-item.active	Item de menu ativo com borda esquerda azul e background semitransparente
mirror-overlay.open	Modal do espelho: display:flex ao receber a classe .open

4. JavaScript — Geração de Dados (5.030 Colaboradores)

Todos os colaboradores são gerados por uma função determinística baseada em **Linear Congruential Generator (LCG)**. O mesmo seed sempre produz os mesmos dados, garantindo consistência entre sessões.

Gerador Pseudo-Aleatório (LCG)

```
function rng(seed) {  
  let s = seed;  
  return () => {  
    s = (s * 1664525 + 1013904223) & 0xffffffff;  
    return (s >>> 0) / 0xffffffff; // retorna float [0, 1)  
  };  
}  
  
// Uso: cada colaborador tem seu próprio gerador isolado  
const r = rng(42 + idx * 7919); // seed único por índice
```

Estrutura do Colaborador (genEmp)

```
function genEmp(idx) {  
  const r = rng(42 + idx * 7919);  
  // Gera nome único (first + middle + sobrenome)  
  // Seleciona setor aleatório e cargo compatível com o setor  
  // Gera CPF e PIS com dígitos verificadores corretos  
  // Define situação: Ativo | Desligado | Férias  
  // Gera status de assinatura (64.4% assinado, resto pendente)  
  return {  
    idx, matricula, nome, cpf, pis,  
    cargo, setor, admissao, tipoSal,  
    situacao, termDate, feriasStart,  
    assinaturaStatus, seed  
  };  
}
```

Validação de CPF Fictício

```
function genCPF(r) {  
  const d = Array.from({length:9}, () => Math.floor(r() * 10));  
  let s1 = d.reduce((a,v,i) => a + v * (10 - i), 0);  
  let d1 = (s1 * 10) % 11; if (d1 >= 10) d1 = 0;  
  let s2 = d.reduce((a,v,i) => a + v * (11 - i), 0) + d1 * 2;  
  let d2 = (s2 * 10) % 11; if (d2 >= 10) d2 = 0;  
  const n = [...d, d1, d2];  
  return `${n.slice(0,3).join("").slice(3,6).join("").slice(6,9).join("")}-${n[9]}${n[10]}`;  
}
```

Desligamentos e Férias

```
// Desligamentos em dias úteis alternados  
function getTermDates(y, m, cnt) {  
  const days = [];  
  for (let d = 2; d <= dimMonth(y, m); d++)  
    if (!isWE(y, m, d)) days.push(d);  
  const result = [];
```

```
for (let i = 0; i < cnt && i*2 < days.length; i++)
result.push(days[i * 2]); // dias alternados

return result;
}

// Maio: 16 desligados | Junho: 14 desligados
const TERM_MAY = getTermDates(2026, 5, 16);
const TERM_JUN = getTermDates(2026, 6, 14);

// Férias: idx 0-9 = início 04/05 | idx 10-19 = início 01/06
if (idx >= 0 && idx < 10) { situacao = "Férias"; feriasStart = {y:2026, m:5, d:4} }
if (idx >= 10 && idx < 20) { situacao = "Férias"; feriasStart = {y:2026, m:6, d:1} }
```


5. JavaScript — Lógica de Ponto (genDayPoint)

A função **genDayPoint(emp, y, m, d, r)** determina o tipo de cada dia e gera as marcações. Ela aplica todas as regras de negócio: desligamento, férias, feriados, fins de semana, faltas, atestados, abonos, esquecimentos e variações de horário.

```
function genDayPoint(emp, y, m, d, r) {
  const dow = new Date(y, m-1, d).getDay(); // 0=Dom, 6=Sáb
  const fKey = `${y}-${fmt2(m)}-${fmt2(d)}`;

  // 1. Sem ponto após desligamento
  if (emp.termDate) {
    const td = emp.termDate;
    if (dia > data_desligamento) return { tipo: "sem-ponto" };
    if (dia === data_desligamento) return { tipo: "desligamento" };
  }

  // 2. Férias (30 dias corridos a partir de feriasStart)
  if (emp.feriasStart) {
    const startTs = new Date(fs.y, fs.m-1, fs.d).getTime();
    const endTs = startTs + 30 * 86400000;
    if (dayTs >= startTs && dayTs < endTs)
      return { tipo: "ferias" };
  }

  // 3. Fim de semana (Sáb 6% chance de trabalhar)
  if (dow === 0 || dow === 6) {
    if (dow === 6 && r() < 0.07) { /* sabado trabalhado */ }
    else return { dow, tipo: "descanso", obs: dow===6 ? "FOLGA COMPENSATÓRIA" : "DSR" };
  }

  // 4. Feriado nacional
  if (FERIADOS[fKey]) return { tipo: "feriado", obs: FERIADOS[fKey] };

  // 5. Situações controladas (2% falta, 1% atestado, etc.)
  const rv = r();
  if (rv < 0.02) return { tipo: "falta" };
  if (rv < 0.03) return { tipo: "atestado" };
  if (rv < 0.04) return { tipo: "abono" };
  if (rv < 0.055) return { tipo: "esquecimento", el: ... };

  // 6. Dia normal - variações realistas
  const e1 = addMin("08:00", rBetween(r, -8, 15)); // entrada
  const s1 = addMin("12:00", rBetween(r, -5, 12)); // saída almoço
  const e2 = addMin("13:00", rBetween(r, -5, 15)); // retorno almoço
  const s2 = addMin("17:48", s2offset); // saída

  const workMins = (pTime(s1)-pTime(e1)) + (pTime(s2)-pTime(e2));
  const heMins = workMins - 528; // 8h48 = 528 min jornada

  const noturno = pTime(s2) > pTime("22:00") ? ... : 0;
  const atrasaMins = pTime(e1) > pTime("08:10") ? pTime(e1)-480 : 0;

  return { tipo: "normal", el, s1, e2, s2, workMins, heMins, atrasaMins, noturno };
}
```

Sumário Mensal (computeSummary)

```
function computeSummary(data) {
  let he=0, hn=0, fal=0, atr=0, ates=0, abo=0, not=0, fer=0;
  for (const d of data) {
    if (d.tipo === "normal" || d.tipo === "sabado") {
      if (d.heMins > 0) he += d.heMins; // horas extras
    }
  }
}
```

```
else hn += Math.abs(d.heMins); // horas negativas
if (d.atrasaMins > 0) atr += d.atrasaMins;
if (d.noturno > 0) not += d.noturno;
}
if (d.tipo === "falta") fal++;
if (d.tipo === "atestado") ates++;
if (d.tipo === "abono") abo++;
if (d.tipo === "ferias") fer++;
}
return { he, hn, fal, atr, ates, abo, not, fer };
}
```

6. JavaScript — Interface e Navegação

Sistema de Navegação (navigate)

```
function navigate(pg) {
  // Oculta todas as páginas
  document.querySelectorAll(".page").forEach(p => p.classList.remove("active"));
  // Remove active de todos os itens do menu
  document.querySelectorAll(".sidebar-item").forEach(i => i.classList.remove("active"));
  // Ativa a página solicitada
  document.getElementById(`page-${pg}`).classList.add("active");
  // Marca o item de menu correspondente
  document.querySelectorAll(".sidebar-item").forEach(i => {
    if (i.getAttribute("onclick") === `navigate("${pg}")`)
      i.classList.add("active");
  });
  curPage = pg;
  // Renderiza conteúdo sob demanda
  if (pg === "colaboradores") { preload(300); buildTableData(); filterTable(); }
  if (pg === "pendencias") { buildPendData(); renderPendRows(); updatePendKPIs(); }
  if (pg === "exportar") { preload(300); buildExpData(); filterExport(); }
}
```

Filtros e Busca em Tempo Real

```
function filterTable() {
  const q = document.getElementById("searchInput").value.toLowerCase();
  const st = document.getElementById("filterSetor").value;
  const cg = document.getElementById("filterCargo").value;
  const si = document.getElementById("filterSit").value;
  const pe = document.getElementById("filterPend").value;

  filteredData = tableData.filter(e => {
    if (q && !e.nome.toLowerCase().includes(q) && !e.matricula.toLowerCase().includes(q)) return false;
    if (st && e.setor !== st) return false;
    if (cg && e.cargo !== cg) return false;
    if (si && e.situacao !== si) return false;
    if (pe === "Pendente" && e.assinaturaStatus === "Assinado") return false;
    if (pe === "Assinado" && e.assinaturaStatus !== "Assinado") return false;
    return true;
  });
  curTblPage = 1;
  renderTblRows();
}
```

Paginação Genérica

```
function renderPagBtns(containerId, cur, total, fn) {
  let h = `<button class="page-btn" onclick="${fn} (${cur-1})" ${cur<=1?"disabled":""}></button>`;
  const ar = 2; // janela de páginas ao redor da atual
  for (let p = 1; p <= total; p++) {
    if (p===1 || p===total || Math.abs(p-cur) <= ar)
      h += `<button class="page-btn ${p===cur?"active":""}" onclick="${fn} (${p})">${p}</button>`;
  }
}
```

```
else if (Math.abs(p-cur) === ar+1)
h += `<button class="page-btn" disabled>...</button>`;
}
h += `<button class="page-btn" onclick="${fn} (${cur+1})" ${cur>total?"disabled":""}>></button>`;
document.getElementById(containerId).innerHTML = h;
}
```

7. JavaScript — Espelho de Ponto (buildMirrorHTML)

A função **buildMirrorHTML(emp)** constrói o HTML completo do espelho de ponto fiel ao modelo corporativo. Ela gera os dados do mês com **genMonth()**, aplica as regras de cor e ocorrência para cada dia, e monta o rodapé "Resumo Pagamentos" com códigos e horas.

Tipos de linha e regras de exibição

descanso (Sáb)	Linha cinza · Coluna Ocorrência: FOLGA COMPENSATÓRIA
descanso (Dom)	Linha cinza · Coluna Ocorrência: DSR
feriado	Linha amarela · Ocorrência: 00013CORPUSCHRISTI etc. · Coluna Horas: 07:20
ferias	Linha verde · Ocorrência: 00013FÉRIAS · Coluna Horas: 07:20 (dias alternados)
falta	Linha vermelha · Ocorrência: FALTA MARCAÇÃO
atestado	Linha branca · Ocorrência: 00021ATESTADOS · Abono: 08:48
abono	Linha branca · Ocorrência: 00101ABONO · Abono: 08:48
esquecimento	Linha vermelha · Mostra e1 (entrada) · Ocorrência: FALTA MARCAÇÃO
normal	Linha branca/alternada · Todas as marcações · H.Extra/Neg · Adicional noturno
sabado	Linha normal · Ocorrência: SAB.TRABALHADO · horas extras 100%

Resumo Pagamentos (rodapé fiel ao modelo)

```
// Gera linhas do Resumo Pagamentos com código, descrição, referência e horas
const resumoRows = [
sum.fer > 0 ? `00006 | FÉRIAS DIAS | ${sum.fer} | ${fmtH(sum.fer*528)}` : "",
sum.ates > 0 ? `00019 | ATESTADOS | ${ates*8.8} | ${fmtH(ates*528)}` : "",
sum.fal > 0 ? `00022 | DESC. FALTAS E ATRASOS | {...} | {...}` : "",
`00086 | EXTRAS DOM E FER | 5.50 | 05:30`,
sum.not > 0 ? `00092 | EXTRAS 100% NOTURNAS | ${not/60} | ${fmtH(not)}` : "",
he100 > 0 ? `00096 | EXTRAS 100% | ${he100/60} | ${fmtH(he100)}` : "",
he50 > 0 ? `00099 | EXTRAS 50% | ${he50/60} | ${fmtH(he50)}` : "",
].filter(Boolean);

// Cálculo de extras 50% e 100%:
const he50 = Math.floor(summary.he * 0.6); // 60% das extras = 50%
const he100 = summary.he - he50; // restante = 100%
```

8. JavaScript — Exportação PDF e CSV

Exportação PDF individual e em lote

```
// Exporta um único espelho em PDF (abre janela de impressão)
function exportOnePDF(idx) {
  openPrintWindow(
    buildMirrorHTML(EMPLOYEES[idx]),
    `Espelho - ${EMPLOYEES[idx].nome} - ${curCompLabel}`
  );
}

// Exporta múltiplos espelhos em um único PDF
function expExportPDF() {
  const sel = expData.filter(e => expSelected.has(e.idx));
  if (sel.length === 1) { exportOnePDF(sel[0].idx); return; }
  // Concatena HTMLs com page-break entre cada espelho
  const htmls = sel
    .map(e => buildMirrorHTML(e))
    .join('<div style="page-break-after:always"></div>');
  openPrintWindow(htmls, `Espelhos - ${curCompLabel} (${sel.length})`);
}

// Abre janela com CSS otimizado para impressão A4
function openPrintWindow(html, title) {
  const w = window.open("", "_blank", "width=950,height=750");
  w.document.write(`<!DOCTYPE html><html><head>
    <title>${title}</title>
    <style>${PRINT_CSS}</style> /* CSS limpo sem sidebar/topbar */
  </head><body>${html}</body></html>`);
  w.document.close();
  setTimeout(() => w.print(), 600); // aguarda renderização
}
```

Exportação CSV

```
function expExportCSV() {
  const sel = expData.filter(e => expSelected.has(e.idx));
  const rows = [
    ["Matrícula", "Nome", "Cargo", "Setor", "Situação",
    "Competência", "H.Extras", "H.Negativas", "Status"]
  ];
  sel.forEach(e => rows.push([
    e.matricula, e.nome, e.cargo, e.setor, e.situacao,
    curCompLabel, fmtH(e.he), fmtH(e.hn), e.assinaturaStatus
  ]));
  // BOM UTF-8 (\u000b) para Excel reconhecer acentos
  const csv = rows.map(r => r.map(v => `"${v}"`).join(",")).join("\n");
  const blob = new Blob(["\u000b" + csv], { type: "text/csv;charset=utf-8" });
  const a = document.createElement("a");
  a.href = URL.createObjectURL(blob);
  a.download = `espelhos-${curComp}.csv`;
  a.click();
}
```

9. Regras de Negócio Implementadas

Desligamentos	Maio: 16 colaboradores (idx 5000-5015) · Junho: 14 (idx 5016-5029). Todos em dias úteis, distribuídos em dias alternados (days[*2]). Sem marcações após a data.
Férias	Grupo A (idx 0-9): início 04/05/2026, 30 dias corridos. Grupo B (idx 10-19): início 01/06/2026, 30 dias. Espelho exibe 00013FÉRIAS e horas DSR alternadas.
Feriados	Dia do Trabalho 01/05, Corpus Christi 21/05 e 11/06. Exibidos em amarelo com código de ocorrência e abono de 07:20.
Fins de semana	Sábado: FOLGA COMPENSATÓRIA · Domingo: DSR. 7% de chance de sábado trabalhado (com horas extras 100%).
Ocorrências	2% falta injustificada · 1% atestado médico · 1% abono · 1,5% esquecimento de marcação. Distribuição controlada por seed.
Horas extras	Calculadas como workMins - 528 (8h48). 60% classificadas como 50%, 40% como 100%. Adicional noturno para saídas após 22h.
Atrasos	Detectados quando pTime(e1) > pTime("08:10"). Valor em minutos armazenado em atrasaMins. Exibe "FALTA MARCAÇÃO" na coluna Ocorrência.
Status espelho	64,4% Assinado · 13,6% Pend. Colaborador · 15% Pend. Gestor · 7% Pend. RH. Determinado por seed do colaborador.
Cache de sumários	summaryCache['\${idx}-\${comp}'] evita recalculer o mês completo a cada re-render. Pré-carrega primeiros 200-300 colaboradores ao navegar.
Matrícula	Formato C00001 a C05030. Gerada como `C\${String(idx+1).padStart(5,"0")}`. Nunca repetida.

10. Código Fonte Integral (vantech-ponto.html)

■ O arquivo possui 82,692 bytes e 1,013 linhas. A seguir está o código completo dividido por seção.

CSS Completo

Variáveis e Base

```
:root(--brand:#1a56db;--brand-dark:#1245b8;--sidebar-bg:#0f172a;--sidebar-text:#94a3b8;--surface:#fff;--surface-2:#f8fafc;--border:#e2e8f0;--text:#0f172a;--text-2:#64748b;--success:#10b981;--warning:#f59e0b;--danger:#ef4444;--radius:8px;--shadow:0 1px 3px rgba(0,0,0,.08),0 1px 2px rgba(0,0,0,.06);--shadow-lg:0 10px 15px rgba(0,0,0,.08),0 4px 6px rgba(0,0,0,.05)){*{box-sizing:border-box;margin:0;padding:0}body{font-family:'Segoe UI',system-ui,sans-serif;background:#f1f5f9;color:var(--text);display:flex;min-height:100vh;font-size:13px}
```

Sidebar

```
#sidebar{width:240px;min-height:100vh;background:var(--sidebar-bg);display:flex;flex-direction:column;position:fixed;left:0;top:0;bottom:0;z-index:100}.sidebar-logo{padding:20px 16px;border-bottom:1px solid #1e293b}.logo-mark{display:flex;align-items:center;gap:10px}.logo-icon{width:32px;height:32px;background:var(--brand);border-radius:8px;display:flex;align-items:center;justify-content:center;font-weight:800;color:#fff;font-size:14px}.logo-name{font-weight:700;color:#fff;font-size:14px;line-height:1.2}.logo-sub{color:var(--sidebar-text);font-size:10px}.sidebar-section{padding:20px 0 8px}.sidebar-label{color:#475569;font-size:9px;font-weight:700;letter-spacing:1px;text-transform:uppercase;padding:0 16px 8px}.sidebar-item{display:flex;align-items:center;gap:10px;padding:9px 16px;color:var(--sidebar-text);cursor:pointer;border-left:3px solid transparent;transition:all .15s;font-size:13px}.sidebar-item:hover{color:#cbd5e1;background:rgba(255,255,255,.04)}.sidebar-item.active{color:#fff;background:rgba(26,86,219,.15);border-left-color:var(--brand)}.sidebar-item svg{width:16px;height:16px;flex-shrink:0}.sidebar-badge{margin-left:auto;background:var(--brand);color:#fff;font-size:9px;padding:2px 6px;border-radius:10px;font-weight:700}.sidebar-footer{margin-top:auto;padding:12px 16px;border-top:1px solid #1e293b;display:flex;align-items:center;gap:10px}.user-avatar{width:30px;height:30px;background:var(--brand);border-radius:50%;display:flex;align-items:center;justify-content:center;color:#fff;font-size:11px;font-weight:700;flex-shrink:0}.user-info{font-size:11px;color:#cbd5e1;line-height:1.3}.user-info span{display:block;color:#94a3b8;font-size:10px}
```

Main

```
#main{margin-left:240px;flex:1;display:flex;flex-direction:column;min-height:100vh}.topbar{background:#fff;border-bottom:1px solid var(--border);padding:0 24px;height:56px;display:flex;align-items:center;gap:16px;position:sticky;top:0;z-index:50}.topbar-title{font-size:15px;font-weight:600}.topbar-sep{color:var(--border)}.topbar-sub{color:var(--text-2);font-size:13px}.topbar-right{margin-left:auto;display:flex;align-items:center;gap:12px}.comp-selector{display:flex;align-items:center;background:var(--surface-2);border:1px solid var(--border);border-radius:var(--radius);overflow:hidden}.comp-btn{padding:6px 14px;font-size:12px;font-weight:500;cursor:pointer;color:var(--text-2);background:none;border:none;transition:all .15s}
```



```
.comp-btn.active{background:var(--brand);color:#fff}
.icon-btn(width:34px;height:34px;border:1px solid var(--border);border-radius:var(--radius);background:none;cursor:pointer;display:flex;align-items:center;justify-content:center;color:var(--text-2) }
```

Content

```
.content{padding:24px;flex:1}
.page{display:none}
.page.active{display:block}
.page-header{margin-bottom:20px}
.page-title{font-size:20px;font-weight:700}
.page-desc(color:var(--text-2);font-size:13px;margin-top:2px)
```

Cards

```
.cards-grid{display:grid;grid-template-columns:repeat(4,1fr);gap:16px;margin-bottom:24px}
.card{background:var(--surface);border:1px solid var(--border);border-radius:var(--radius);padding:16px;box-shadow:var(--shadow) }
.card-header{display:flex;align-items:flex-start;justify-content:space-between;margin-bottom:12px}
.card-label{font-size:11px;font-weight:600;color:var(--text-2);text-transform:uppercase;letter-spacing:.5px}
.card-icon(width:36px;height:36px;border-radius:8px;display:flex;align-items:center;justify-content:center)
.card-icon svg{width:18px;height:18px}
.icon-blue(background:#dbeafe;color:#1d4ed8).icon-green(background:#d1fae5;color:#059669).icon-red(background:#fee2e2;color:#dc2626).icon-yellow(background:#fff3cd;color:#ffc107).icon-purple(background:#e2e3e5;color:#6c757d)
.card-value{font-size:26px;font-weight:700;line-height:1}
.card-delta{font-size:11px;margin-top:6px;color:var(--text-2) }
.delta-up(color:var(--success)).color-danger(color:var(--danger)).color-warning(color:var(--warning)).color-success(color:var(--success))
```

Charts

```
.charts-grid{display:grid;grid-template-columns:repeat(3,1fr);gap:16px;margin-bottom:24px}
.chart-card{background:var(--surface);border:1px solid var(--border);border-radius:var(--radius);padding:16px;box-shadow:var(--shadow) }
.chart-card.span2{grid-column:span 2}
.chart-title{font-size:13px;font-weight:600;margin-bottom:4px}
.chart-sub{font-size:11px;color:var(--text-2);margin-bottom:14px}
```

Table

```
.table-card{background:var(--surface);border:1px solid var(--border);border-radius:var(--radius);box-shadow:var(--shadow);overflow:hidden}
.table-toolbar{padding:14px 16px;display:flex;align-items:center;gap:10px;border-bottom:1px solid var(--border);flex-wrap:wrap}
.table-toolbar-title{font-size:14px;font-weight:600;white-space:nowrap}
.search-box{display:flex;align-items:center;gap:8px;border:1px solid var(--border);border-radius:var(--radius);padding:7px 12px;background:var(--surface-2);flex:1;max-width:320px}
.search-box input{border:none;background:none;outline:none;font-size:13px;color:var(--text);width:100%}
.search-box svg{width:15px;height:15px;color:var(--text-2);flex-shrink:0}
.filter-select(border:1px solid var(--border);border-radius:var(--radius);padding:7px 10px;font-size:12px;background:var(--surface);color:var(--text);outline:none;cursor:pointer)
.btn{display:inline-flex;align-items:center;gap:6px;padding:7px 14px;border-radius:var(--radius);font-size:12px;font-weight:500;cursor:pointer;border:none;transition:all .15s;white-space:nowrap}
.btn-primary{background:var(--brand);color:#fff}.btn-primary:hover{background:var(--brand-dark) }
.btn-secondary{background:var(--surface-2);border:1px solid var(--border);color:var(--text)}.btn-secondary:hover{background:#e2e8f0}
.btn-danger{background:#fee2e2;border:1px solid #fca5a5;color:#dc2626}
.btn svg{width:14px;height:14px}
.btn.disabled{opacity:.4;cursor:not-allowed}
.table-wrap{overflow-x:auto}
table{width:100%;border-collapse:collapse}
```

```

th{background:var(--surface-2);font-size:11px;font-weight:700;color:var(--text-2);text-transform:uppercase;letter-spacing:.5px;padding:10px 14px;text-align:left;border-bottom:1px solid var(--border);white-space:nowrap;cursor:pointer;user-select:none}
th:hover{background:#e9edf2}
td{padding:10px 14px;border-bottom:1px solid var(--border);font-size:12px;color:var(--text);white-space:nowrap}
tr:last-child td{border-bottom:none}
tr:hover td{background:#f8fafc}
.badge{display:inline-flex;align-items:center;gap:4px;padding:3px 8px;border-radius:10px;font-size:10px;font-weight:600}
.badge-green{background:#d1fae5;color:#065f46}.badge-red{background:#fee2e2;color:#991b1b}.badge-yellow{background:#fef3c7;color:#92400e}.badge-blue{background:#dbeeaf;color:#1e40af}.badge-gray{background:#f1f5f9;color:#475569}
.badge-dot{width:5px;height:5px;border-radius:50%;background:currentColor}
.mat-code{font-family:monospace;font-size:11px;color:var(--brand);font-weight:600}
.pagination{display:flex;align-items:center;justify-content:space-between;padding:12px 16px;border-top:1px solid var(--border)}
.pagination-info{font-size:12px;color:var(--text-2)}
.pagination-btns{display:flex;gap:4px}
.page-btn{width:30px;height:30px;border:1px solid var(--border);border-radius:6px;background:none;cursor:pointer;display:flex;align-items:center;justify-content:center;font-size:12px;color:var(--text-2);transition:all .15s}
.page-btn.active{background:var(--brand);color:#fff;border-color:var(--brand)}
.page-btn:hover:not(.active):not(:disabled){background:var(--surface-2)}
.page-btn.disabled{opacity:.4;cursor:default}

```

Info strip

```

.info-strip{background:#eff6ff;border:1px solid #bfdbfef;border-radius:var(--radius);padding:10px 14px;font-size:12px;color:#1e40af;display:flex;align-items:center;gap:8px;margin-bottom:16px}
.info-strip svg{width:14px;height:14px;flex-shrink:0}

```

KPI row

```

.kpi-row{display:flex;gap:12px;margin-bottom:16px;flex-wrap:wrap}
.kpi{background:#fff;border:1px solid var(--border);border-radius:var(--radius);padding:12px 16px;flex:1;min-width:140px}
.kpi-label{font-size:10px;font-weight:700;color:var(--text-2);text-transform:uppercase;letter-spacing:.5px}
.kpi-val{font-size:22px;font-weight:700;margin-top:2px}
.kpi-sub{font-size:10px;color:var(--text-2)}

```

Modal

```

#mirror-overlay{display:none;position:fixed;inset:0;background:rgba(0,0,0,.55);z-index:200;align-items:center;justify-content:center;padding:20px}
#mirror-overlay.open{display:flex}
#mirror-modal{background:#fff;border-radius:10px;width:100%;max-width:980px;max-height:92vh;display:flex;flex-direction:column;box-shadow:var(--shadow-1g)}
.modal-header{padding:16px 20px;border-bottom:1px solid var(--border);display:flex;align-items:center;justify-content:space-between;flex-wrap:wrap;gap:8px}
.modal-title{font-size:15px;font-weight:700}
.modal-actions{display:flex;gap:8px}
.modal-body{overflow-y:auto;flex:1;padding:20px;background:#f1f5f9}

```

===== ESPELHO DE PONTO (fiel ao modelo insoft4) =====

```

.ep-wrap{background:#fff;font-family:Arial,Helvetica,sans-serif;font-size:9px;width:100%;border:1px solid #888;padding:6px}
.ep-header{border:1px solid #999;margin-bottom:2px}
.ep-logo-row{display:flex;align-items:stretch;border-bottom:1px solid #999}
.ep-logo-box{background:#1a3f7a;display:flex;align-items:center;justify-content:center;padding:6px 10px;min-width:90px}
.ep-logo-text{color:#fff;font-size:11px;font-weight:800;line-height:1.1;text-align:center}

```

```

.ep-logo-sub{color:#a0c4ff;font-size:7px;font-weight:400}
.ep-title-box{flex:1;display:flex;flex-direction:column;align-items:center;justify-content:center;padding:4px}
.ep-title{font-size:13px;font-weight:700;color:#000;text-align:center}
.ep-comp-line{font-size:9px;color:#333;text-align:center;margin-top:2px}
.ep-page-box{text-align:right;padding:4px 8px;font-size:9px;color:#333;white-space:nowrap}
.ep-empresa-row{padding:4px 8px;font-size:9px;font-weight:700;color:#000;border-bottom:1px solid #ccc}
.ep-cnpj-row{padding:3px 8px;font-size:8px;color:#333;display:flex;justify-content:space-between;border-bottom:1px solid #ccc}
.ep-fields{display:grid;grid-template-columns:auto 1fr auto 1fr;border-top:1px solid #ccc}
.ep-field{display:contents}
.ep-fl{font-size:7px;font-weight:700;color:#333;text-transform:uppercase;padding:2px 6px;border-right:1px solid #ccc;border-bottom:1px solid #ccc;white-space:nowrap;background:#f5f5f5}
.ep-fv{font-size:9px;font-weight:600;color:#000;padding:2px 6px;border-right:1px solid #ccc;border-bottom:1px solid #ccc}
.ep-fields2{display:grid;grid-template-columns:auto 1fr auto 1fr auto 1fr;border-top:0}
.ep-name-row{display:flex;border-top:0;border-bottom:1px solid #ccc}
.ep-name-l{font-size:7px;font-weight:700;color:#333;text-transform:uppercase;padding:2px 6px;border-right:1px solid #ccc;background:#f5f5f5;white-space:nowrap}
.ep-name-v{font-size:9px;font-weight:700;color:#000;padding:2px 6px;flex:1;border-right:1px solid #ccc}
.ep-pis-l{font-size:7px;font-weight:700;color:#333;text-transform:uppercase;padding:2px 6px;border-right:1px solid #ccc;background:#f5f5f5;white-space:nowrap}
.ep-pis-v{font-size:9px;font-weight:600;color:#000;padding:2px 6px}

```

tabela de ponto

```

.ep-table{width:100%;border-collapse:collapse;margin-top:3px;font-size:7.5px}
.ep-table th{background:#1a3f7a;color:#fff;font-size:7px;font-weight:700;padding:3px 3px;text-align:center;border:1px solid #0f2a5a}
.ep-table th.th-jornada{background:#2a5298}
.ep-table th.th-marc{background:#1a3f7a}
.ep-table th.th-total{background:#1a3f7a}
.ep-table td{font-size:7.5px;padding:2px 3px;text-align:center;border:1px solid #ccc;color:#000}
.ep-table tr.we td{background:#eeeeee;color:#888}
.ep-table tr.fer td{background:#fff7e0}
.ep-table tr.ferias-row td{background:#e8f5e9;font-style:italic}
.ep-table tr.falta-row td{background:#fff0f0}
.ep-table tr.even td{background:#f7f9fc}
.ep-table td.oc{text-align:left;font-size:7px}

```

resumo pagamentos

```

.ep-resumo{margin-top:4px;border:1px solid #999}
.ep-resumo-title{background:#1a3f7a;color:#fff;font-size:8px;font-weight:700;padding:3px 6px;display:grid;grid-template-columns:60px 1fr 80px 80px;gap:0}
.ep-resumo-row{display:grid;grid-template-columns:60px 1fr 80px 80px;border-top:1px solid #ddd;font-size:8px}
.ep-resumo-row div{padding:2px 6px;border-right:1px solid #ddd}
.ep-resumo-row div:last-child{border-right:none}
.ep-total-row{display:grid;grid-template-columns:1fr auto auto;border-top:2px solid #999;font-size:8px;font-weight:700;background:#f5f5f5}
.ep-total-row div{padding:3px 6px;border-right:1px solid #ddd}
.ep-sign{margin-top:20px;display:flex;justify-content:center}
.ep-sign-line{text-align:center}
.ep-sign-rule{border-top:1px solid #000;width:280px;margin:0 auto 4px}
.ep-sign-label{font-size:8px;color:#333;text-align:center}

```

Export page

```

.export-panel{background:#fff;border:1px solid var(--border);border-radius:var(--radius);box-shadow:var(--shadow);overflow:hidden;margin-bottom:20px}
.export-search-area{padding:20px}

```

```
.export-search-title{font-size:14px;font-weight:700;margin-bottom:4px}
.export-search-desc{font-size:12px;color:var(--text-2);margin-bottom:16px}
.export-fields{display:flex;gap:10px;flex-wrap:wrap;align-items:flex-end}
.export-field{display:flex;flex-direction:column;gap:6px}
.export-label{font-size:10px;font-weight:700;color:var(--text-2);text-transform:uppercase;letter-spacing:.5px}
.export-results-bar{padding:10px 16px;background:#fafbfc;border-top:1px solid var(--border);border-bottom:1px solid var(--border);display:flex;align-items:center;justify-content:space-between;flex-wrap:wrap;gap:8px}
```

Jornadas / Parametros

```
.params-grid{display:grid;grid-template-columns:1fr 1fr;gap:16px}
.param-card{background:#fff;border:1px solid var(--border);border-radius:var(--radius);padding:16px;box-shadow:var(--shadow)}
.param-card-title{font-size:12px;font-weight:700;color:var(--text-2);text-transform:uppercase;letter-spacing:.5px;margin-bottom:12px;padding-bottom:8px;border-bottom:1px solid var(--border)}
.param-row{display:flex;justify-content:space-between;align-items:center;padding:7px 0;border-bottom:1px solid var(--border);font-size:12px}
.param-row:last-child{border-bottom:none}
.param-key{color:var(--text-2)}
.param-val{font-weight:600;color:var(--text)}
.fw-600{font-weight:600}
.text-right{text-align:right}
```

Scrollbar

```
::-webkit-scrollbar{width:6px;height:6px}
::-webkit-scrollbar-track{background:transparent}
::-webkit-scrollbar-thumb{background:#cbd5e1;border-radius:3px}
```

HTML — Sidebar e Topbar

```
<!-- SIDEBAR -->
<nav id="sidebar">
<div class="sidebar-logo">
<div class="logo-mark">
<div class="logo-icon">VT</div>
<div>
<div class="logo-name">VanTech</div>
<div class="logo-sub">Controle de Ponto · RH</div>
</div>
</div>
</div>
<div class="sidebar-section">
<div class="sidebar-label">Principal</div>
<div class="sidebar-item active" onclick="navigate('dashboard')">
<svg fill="none" stroke="currentColor" viewBox="0 0 24 24" stroke-width="2"><rect x="3" y="3" width="7" height="7" rx="1"/><rect x="14" y="3" width="7" height="7" rx="1"/><rect x="3" y="14" width="7" height="7" rx="1"/><rect x="14" y="14" width="7" height="7" rx="1"/></svg>
Dashboard
</div>
<div class="sidebar-item" onclick="navigate('colaboradores')">
<svg fill="none" stroke="currentColor" viewBox="0 0 24 24" stroke-width="2"><path d="M17 21v-2a4 4 0 0 0-4-4H5a4 4 0 0 0-4 4v2"/><circle cx="9" cy="7" r="4"/><path d="M23 21v-2a4 4 0 0 0-3-3.87M16 3.13a4 4 0 0 1 0 7.75"/></svg>
Colaboradores
<span class="sidebar-badge">5.030</span>
</div>
<div class="sidebar-item" onclick="navigate('pendencias')">
```

```

<svg fill="none" stroke="currentColor" viewBox="0 0 24 24" stroke-width="2"><path d="M10.29 3.86 1.82 18a2 2 0 0
0 1.71 3h16.94a2 2 0 0 0 1.71-3L13.71 3.86a2 2 0 0 0-3.42 0z"/><line x1="12" y1="9" x2="12" y2="13"/><line
x1="12" y1="17" x2="12.01" y2="17"/></svg>
Pendências
<span class="sidebar-badge">347</span>
</div>
</div>
<div class="sidebar-section">
<div class="sidebar-label">Relatórios</div>
<div class="sidebar-item" onclick="navigate('exportar')">
<svg fill="none" stroke="currentColor" viewBox="0 0 24 24" stroke-width="2"><path d="M14 2H6a2 2 0 0 0-2 2v16a2
2 0 0 0 2 2h12a2 2 0 0 0 2-2V8z"/><polyline points="14 2 14 8 20 8"/><line x1="16" y1="13" x2="8" y2="13"/><line
x1="16" y1="17" x2="8" y2="17"/></svg>
Exportar Espelhos
</div>
</div>
<div class="sidebar-section">
<div class="sidebar-label">Configuração</div>
<div class="sidebar-item" onclick="navigate('jornadas')">
<svg fill="none" stroke="currentColor" viewBox="0 0 24 24" stroke-width="2"><circle cx="12" cy="12"
r="10"/><polyline points="12 6 12 12 16 14"/></svg>
Jornadas
</div>
<div class="sidebar-item" onclick="navigate('parametros')">
<svg fill="none" stroke="currentColor" viewBox="0 0 24 24" stroke-width="2"><circle cx="12" cy="12" r="3"/><path
d="M19.07 4.93a10 10 0 0 1 0 14.14M4.93 4.93a10 10 0 0 0 0 14.14"/></svg>
Parâmetros
</div>
</div>
<div class="sidebar-footer">
<div class="user-avatar">RH</div>
<div class="user-info">Administrador RH<span>rh@vantech.com.br</span></div>
</div>
</nav>
<!-- MAIN -->
<div id="main">
<header class="topbar">
<span class="topbar-title">VanTech - Data Solutions</span>
<span class="topbar-sep">></span>
<span class="topbar-sub" id="page-breadcrumb">Dashboard</span>
<div class="topbar-right">
<div class="comp-selector">

```

HTML — Páginas Principais

```

<button class="comp-btn active" id="btn-maio" onclick="setComp('maio')">Maio/2026</button>
<button class="comp-btn" id="btn-junho" onclick="setComp('junho')">Junho/2026</button>
</div>
<button class="icon-btn"><svg width="16" height="16" fill="none" stroke="currentColor" viewBox="0 0 24 24"
stroke-width="2"><path d="M18 8A6 6 0 0 0 6 8c0 7-3 9-3 9h18s-3-2-3-9"/><path d="M13.73 21a2 2 0 0 1-3.46
0"/></svg></button>
</div>
</header>
<div class="content">
<!-- DASHBOARD -->
<div id="page-dashboard" class="page active">

```

```

<div class="page-header">
<div class="page-title">Dashboard Executivo</div>
<div class="page-desc">Visão geral · Competência: <span id="dash-comp-label">Maio/2026</span></div>
</div>
<div class="cards-grid">
<div class="card"><div class="card-header"><div class="card-label">Total Colaboradores</div><div
class="card-icon icon-blue"><svg fill="none" stroke="currentColor" viewBox="0 0 24 24" stroke-width="2"><path
d="M17 21v-2a4 4 0 0 0-4-4H5a4 4 0 0 0-4 4v2"/><circle cx="9" cy="7" r="4"/></svg></div></div><div
class="card-value">5.030</div><div class="card-delta">Registro completo</div></div>
<div class="card"><div class="card-header"><div class="card-label">Ativos</div><div class="card-icon
icon-green"><svg fill="none" stroke="currentColor" viewBox="0 0 24 24" stroke-width="2"><path d="M22 11.08V12a10
10 0 1 1-5.93-9.14"/><polyline points="22 4 12 14.01 9 11.01"/></svg></div></div><div
class="card-value">5.000</div><div class="card-delta delta-up">↑ dentro do mês</div></div>
<div class="card"><div class="card-header"><div class="card-label">Desligados</div><div class="card-icon
icon-red"><svg fill="none" stroke="currentColor" viewBox="0 0 24 24" stroke-width="2"><circle cx="12" cy="12"
r="10"/><line x1="15" y1="9" x2="9" y2="15"/><line x1="9" y1="9" x2="15" y2="15"/></svg></div></div><div
class="card-value" id="card-desl">16</div><div class="card-delta color-danger">neste período</div></div>
<div class="card"><div class="card-header"><div class="card-label">Em Férias</div><div class="card-icon
icon-yellow"><svg fill="none" stroke="currentColor" viewBox="0 0 24 24" stroke-width="2"><rect x="3" y="4"
width="18" height="18" rx="2"/><line x1="16" y1="2" x2="16" y2="6"/><line x1="8" y1="2" x2="8" y2="6"/><line
x1="3" y1="10" x2="21" y2="10"/></svg></div></div><div class="card-value" id="card-ferias">10</div><div
class="card-delta color-warning">colaboradores</div></div>
<div class="card"><div class="card-header"><div class="card-label">Horas Extras</div><div class="card-icon
icon-purple"><svg fill="none" stroke="currentColor" viewBox="0 0 24 24" stroke-width="2"><circle cx="12" cy="12"
r="10"/><polyline points="12 6 12 16 14"/></svg></div></div><div class="card-value"
id="card-he">1.847h</div><div class="card-delta">total acumulado</div></div>
<div class="card"><div class="card-header"><div class="card-label">Horas Negativas</div><div class="card-icon
icon-red"><svg fill="none" stroke="currentColor" viewBox="0 0 24 24" stroke-width="2"><circle cx="12" cy="12"
r="10"/><line x1="8" y1="12" x2="16" y2="12"/></svg></div></div><div class="card-value"
id="card-hn">389h</div><div class="card-delta color-danger">déficit</div></div>
<div class="card"><div class="card-header"><div class="card-label">Espelhos Assinados</div><div
class="card-icon icon-green"><svg fill="none" stroke="currentColor" viewBox="0 0 24 24" stroke-width="2"><path
d="M14 2H6a2 2 0 0 0-2 2v16a2 2 0 0 0 2 2h12a2 2 0 0 0 2-2V8z"/><polyline points="14 2 14 8 20 8"/><line x1="16"
y1="13" x2="8" y2="13"/></svg></div></div><div class="card-value" id="card-ass">3.241</div><div
class="card-delta delta-up">64,4%</div></div>
<div class="card"><div class="card-header"><div class="card-label">Espelhos Pendentes</div><div
class="card-icon icon-yellow"><svg fill="none" stroke="currentColor" viewBox="0 0 24 24" stroke-width="2"><path
d="M10.29 3.86 1.82 18a2 2 0 0 0 1.71 3h16.94a2 2 0 0 0 1.71-3L13.71 3.86a2 2 0 0 0-3.42 0z"/><line x1="12"
y1="9" x2="12" y2="13"/></svg></div></div><div class="card-value" id="card-pend">1.789</div><div
class="card-delta color-warning">aguardando</div></div>
</div>
<div class="charts-grid">
<div class="chart-card span2"><div class="chart-title">Horas Extras por Setor</div><div class="chart-sub">Total
acumulado no período</div><div style="height:200px;position:relative"><canvas
id="chartSetores"></canvas></div></div>
<div class="chart-card"><div class="chart-title">Tipos de Ocorrência</div><div class="chart-sub">Distribuição no
período</div><div style="height:200px;position:relative"><canvas id="chartOcorrencias"></canvas></div></div>
</div>
<div class="charts-grid">
<div class="chart-card"><div class="chart-title">Horas Extras por Dia</div><div class="chart-sub">Acumulado
diário</div><div style="height:180px;position:relative"><canvas id="chartDiario"></canvas></div></div>
<div class="chart-card"><div class="chart-title">Faltas por Semana</div><div class="chart-sub">No
período</div><div style="height:180px;position:relative"><canvas id="chartFaltas"></canvas></div></div>
<div class="chart-card"><div class="chart-title">Distribuição por Cargo</div><div
class="chart-sub">Percentual</div><div style="height:180px;position:relative"><canvas
id="chartCargos"></canvas></div></div>
</div>
<div class="charts-grid">

```

```

<div class="chart-card span2"><div class="chart-title">Top 10 – Mais Horas Extras</div><div
class="chart-sub">Ranking individual</div><div style="height:220px;position:relative"><canvas
id="chartTop10"></canvas></div></div>
<div class="chart-card"><div class="chart-title">Pendências de Assinatura</div><div class="chart-sub">Status dos
espelhos</div><div style="height:220px;position:relative"><canvas id="chartPendencias"></canvas></div></div>
</div>
</div>
<!-- COLABORADORES -->
<div id="page-colaboradores" class="page">
<div class="page-header">
<div class="page-title">Colaboradores</div>
<div class="page-desc">Espelhos de ponto · <span id="colab-comp-label">Maio/2026</span></div>
</div>
<div class="table-card">
<div class="table-toolbar">
<span class="table-toolbar-title">Espelho de Ponto</span>
<div class="search-box"><svg fill="none" stroke="currentColor" viewBox="0 0 24 24" stroke-width="2"><circle
cx="11" cy="11" r="8"/><line x1="21" y1="21" x2="16.65" y2="16.65"/></svg><input type="text" id="searchInput"
placeholder="Nome ou matrícula..." oninput="filterTable()"></div>
<select class="filter-select" id="filterSetor" onchange="filterTable()"><option value="">Todos os
setores</option></select>
<select class="filter-select" id="filterCargo" onchange="filterTable()"><option value="">Todos os
cargos</option></select>
<select class="filter-select" id="filterSit" onchange="filterTable()"><option value="">Todas
situações</option><option>Ativo</option><option>Desligado</option><option>Férias</option></select>
<select class="filter-select" id="filterPend" onchange="filterTable()"><option value="">Todas
pendências</option><option value="Pendente">Com pendência</option><option value="Assinado">Sem
pendência</option></select>
<button class="btn btn-secondary" onclick="exportCSV()"><svg fill="none" stroke="currentColor" viewBox="0 0 24
24" stroke-width="2"><path d="M21 15v4a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2v-4"/></svg><polyline points="7 10 12 15 17
10"/><line x1="12" y1="15" x2="12" y2="3"/></svg>CSV</button>
</div>
<div class="table-wrap">
<table>
<thead><tr>
<th onclick="sortTable('matricula')">Matricula ▀</th>
<th onclick="sortTable('nome')">Nome ▀</th>
<th onclick="sortTable('cargo')">Cargo ▀</th>
<th onclick="sortTable('setor')">Setor ▀</th>
<th onclick="sortTable('situacao')">Situação ▀</th>
<th onclick="sortTable('he')" class="text-right">H. Extras ▀</th>
<th onclick="sortTable('hn')" class="text-right">H. Neg. ▀</th>
<th>Status Espelho</th>
<th>Ações</th>
</tr></thead>
<tbody id="tbody"></tbody>
</table>
</div>
<div class="pagination"><div class="pagination-info" id="pag-info"></div><div class="pagination-btns"
id="pag-btns"></div></div>
</div>
</div>
<!-- PENDÊNCIAS -->
<div id="page-pendencias" class="page">

```

```
<div class="page-header"><div class="page-title">Pendências</div><div class="page-desc">Espelhos aguardando
assinatura e aprovação</div></div>

<div class="info-strip"><svg fill="none" stroke="currentColor" viewBox="0 0 24 24" stroke-width="2"><circle
cx="12" cy="12" r="10"/><line x1="12" y1="8" x2="12" y2="12"/><line x1="12" y1="16" x2="12.01"
y2="16"/></svg>Existem <strong id="pend-count">1.789</strong> espelhos aguardando ação neste período.</div>

<div class="kpi-row">
<div class="kpi"><div class="kpi-label">Pend. Colaborador</div><div class="kpi-val"
id="pend-colab">1.102</div><div class="kpi-sub">aguardando assinatura</div></div>

<div class="kpi"><div class="kpi-label">Pend. Gestor</div><div class="kpi-val" id="pend-gestor">687</div><div
class="kpi-sub">aguardando aprovação</div></div>

<div class="kpi"><div class="kpi-label">Pend. RH</div><div class="kpi-val" id="pend-rh">347</div><div
class="kpi-sub">aguardando homologação</div></div>

<div class="kpi"><div class="kpi-label">Aprovados RH</div><div class="kpi-val">3.241</div><div
class="kpi-sub">finalizados</div></div>

</div>

<div class="table-card">
<div class="table-toolbar"><span class="table-toolbar-title">Espelhos Pendentes</span>

<div class="search-box"><svg fill="none" stroke="currentColor" viewBox="0 0 24 24" stroke-width="2"><circle
cx="11" cy="11" r="8"/><line x1="21" y1="21" x2="16.65" y2="16.65"/></svg><input type="text" id="searchPend"
placeholder="Buscar..." oninput="filterPendTable()"></div>

</div>

<div class="table-wrap"><table><thead><tr><th>Matricula</th><th>Nome</th><th>Setor</th><th>Cargo</th><th>Status
Collab.</th><th>Status Gestor</th><th>Status RH</th><th>Ações</th></tr></thead><tbody
id="tbody-pend"></tbody></table></div>

<div class="pagination"><div class="pagination-info" id="pag-pend-info"></div><div class="pagination-btns"
id="pag-pend-btns"></div></div>

</div>

</div>

<!-- EXPORTAR ESPELHOS -->

<div id="page-exportar" class="page">
<div class="page-header">

<div class="page-title">Exportar Espelhos de Ponto</div>

<div class="page-desc">Busque colaboradores por matrícula, nome ou setor e exporte os espelhos em PDF ou
CSV</div>

</div>
```

HTML — Modal do Espelho

```
<div id="mirror-overlay">
<div id="mirror-modal">
<div class="modal-header">
<div class="modal-title" id="modal-title">Espelho de Ponto</div>
<div class="modal-actions">

<button class="btn btn-secondary" onclick="printMirror()"><svg fill="none" stroke="currentColor" viewBox="0 0 24
24" stroke-width="2" width="14" height="14"><polyline points="6 9 6 2 18 2 18 9"/><path d="M6 18H4a2 2 0 0
1-2-2v-5a2 2 0 0 1 2-2h16a2 2 0 0 1 2 2v5a2 2 0 0 1-2 2h-2"/><rect x="6" y="14" width="12"
height="8"/></svg>Imprimir / PDF</button>

<button class="btn btn-primary" onclick="closeMirror()">Fechar</button>

</div>

</div>

<div class="modal-body" id="modal-body"></div>

</div>

</div>
```

JavaScript — Dados e Helpers

■ DADOS

```
const NOMES_M=['Lucas','Carlos','Rafael','Anderson','Rodrigo','Felipe','Marcelo','Bruno','Diego','Gabriel','Joã
o','Eduardo','Fernando','Gustavo','Henrique','Leandro','Mateus','Thiago','Vinicius','Allan','André','Caio','Cri
stiano','Daniel','Fábio','Guilherme','Hugo','Iago','Jorge','Kleber','Leonardo','Mário','Natan','Otávio','Patric
k','Renato','Samuel','Tiago','Uliisses','Wagner','Alexsandro','Bernardo','César','Davi','Emerson','Francisco','G
lauber','Humberto','Ivan','José'];
```



```

const NOMES_F=['Amanda','Ana','Beatriz','Carla','Daniela','Elaine','Fernanda','Gabriela','Helena','Isabela','Juliana','Karen','Larissa','Mariana','Natália','Olivia','Patricia','Rafaela','Sandra','Tatiana','Vanessa','Adriana','Barbara','Camila','Debora','Erica','Fatima','Giovana','Heloisa','Ingrid','Jessica','Karine','Leticia','Monica','Nathalia','Paloma','Renata','Sabrina','Tainara','Ursula','Viviane','Aline','Bruna','Claudia','Denise','Edilene','Flavia','Gisele','Herica','Iara'];

const MIDS_M=['Henrique','Eduardo','Augusto','Roberto','Luiz','Carlos','José','Marcos','Ricardo','Paulo','André','Rodrigo','Felipe','Gustavo','Rafael'];

const MIDS_F=['Cristina','Aparecida','Regina','Maria','Lucia','Fernanda','Beatriz','Alexandra','Renata','Claudia','Patricia','Andreia','Vanessa','Simone','Leticia'];

const SOBS=['Silva','Santos','Oliveira','Souza','Rodrigues','Ferreira','Alves','Pereira','Lima','Gomes','Costa','Ribeiro','Martins','Carvalho','Almeida','Lopes','Sousa','Fernandes','Vieira','Barbosa','Rocha','Dias','Monteiro','Cardoso','Nascimento','Moraes','Castro','Correia','Araújo','Machado','Melo','Campos','Pinto','Cruz','Teixeira','Moreira','Ramos','Mendes','Cunha','Freitas','Nunes','Freire','Rezende','Azevedo','Pires','Andrade','Barros','Batista','Cavalcanti','Fonseca'];

const SETORES=['Tecnologia da Informação','Recursos Humanos','Financeiro','Comercial','Marketing','Operações','Logística','Atendimento ao Cliente','Jurídico','Contabilidade','Engenharia','Produção','Qualidade','Compras','Administrativo'];

const CARGOS={'Tecnologia da Informação':['Desenvolvedor Senior','Desenvolvedor Pleno','Analista de Sistemas','Arquiteto de Software','DevOps Engineer','Analista de QA','Scrum Master'],'Recursos Humanos':['Analista de RH','Coordenador de RH','Especialista em T&D','Analista de DP','Gerente de RH'],'Financeiro':['Analista Financeiro','Controller','Contador','Analista de Crédito','Gerente Financeiro','Assistente Financeiro'],'Comercial':['Representante Comercial','Gerente de Vendas','Consultor de Negócios','Coordenador Comercial','Vendedor Senior'],'Marketing':['Analista de Marketing','Designer Gráfico','Gestor de Mídias Sociais','Copywriter','Gerente de Marketing'],'Operações':['Analista de Operações','Supervisor de Operações','Gerente de Operações','Assistente de Operações'],'Logística':['Analista de Logística','Coordenador de Estoque','Assistente de Logística','Gerente de Logística'],'Atendimento ao Cliente':['Atendente','Supervisor de Atendimento','Analista de Suporte','Coordenador de CS'],'Jurídico':['Advogado','Assistente Jurídico','Paralegal','Gerente Jurídico'],'Contabilidade':['Contador','Auxiliar Contábil','Analista Contábil','Supervisor Contábil'],'Engenharia':['Engenheiro Senior','Engenheiro Pleno','Analista de Engenharia','Coordenador de Projetos'],'Produção':['Operador de Produção','Supervisor de Produção','Analista de Produção','Gerente de Produção'],'Qualidade':['Analista de Qualidade','Auditor de Qualidade','Gerente de Qualidade','Inspector de Qualidade'],'Compras':['Analista de Compras','Comprador Senior','Gerente de Compras','Assistente de Compras'],'Administrativo':['Assistente Administrativo','Analista Administrativo','Secretária Executiva','Recepcionista','Gerente Administrativo']};

const FERIADOS={'2026-05-01':'Dia do Trabalho','2026-05-21':'Corpus Christi','2026-06-11':'Corpus Christi'};

function rng(seed){let s=seed;return ()=>{s=(s*1664525+1013904223)&0xffffffff;return(s>>>0)/0xffffffff}}

function fmt2(n){return String(n).padStart(2,'0')}

function fmtH(m){if(m===null||m===undefined)return '-';const s=m<0?'-':'';const a=Math.abs(m);return `${s}${Math.floor(a/60)}:${fmt2(a%60)}`}

function rBetween(r,a,b){return Math.floor(r*(b-a+1))+a}

function isWE(y,m,d){const w=new Date(y,m-1,d).getDay();return w===0||w===6}

function dimMonth(y,m){return new Date(y,m,0).getDate()}

function pTime(hm){if(!hm||hm=='-')return 0;const[h,mi]=hm.split(':').map(Number);return h*60+mi}

function addMin(hm,d){const t=pTime(hm)+d;return`${fmt2(Math.floor(t/60))}:${fmt2(t%60)}`}

function genCPF(r){const d=Array.from({length:9},()=>Math.floor(r()*10));let s1=d.reduce((a,v,i)=>a+v*(10-i),0);let d1=(s1*10)%11;if(d1>=10)d1=0;let s2=d.reduce((a,v,i)=>a+v*(11-i),0)+d1*2;let d2=(s2*10)%11;if(d2>=10)d2=0;const n=[...d,d1,d2];return`${n.slice(0,3).join('').}${n.slice(3,6).join('').}${n.slice(6,9).join('').}-${n[9]}${n[10]}`}

function genPIS(r){const d=Array.from({length:11},()=>Math.floor(r()*10));return`${d.slice(0,3).join('').}${d.slice(3,8).join('').}${d.slice(8,10).join('').}-${d[10]}`}

// Term dates
function getTermDates(y,m,cnt){const days=[];for(let d=2;d<=dimMonth(y,m);d++)if(!isWE(y,m,d))days.push(d);const r=[];for(let i=0;i<cnt&&i*2<days.length;i++)r.push(days[i*2]);return r}

const TERM_MAY=getTermDates(2026,5,16);

const TERM_JUN=getTermDates(2026,6,14);

let EMPLOYEES=[];

const usedNames=new Set();

function genEmp(idx){

```

```

const r= rng(42+idx*7919);
const fem=r()>0.52;
const FN=fem?NOMES_F:NOMES_M,MN=fem?MIDS_F:MIDS_M;
let nome;let tries=0;
do{nome=`${FN[Math.floor(r()*FN.length)]}${MN[Math.floor(r()*MN.length)]}${SOBS[Math.floor(r()*SOBS.length)]}`;tries++;while(usedNames.has(nome)&&tries<20);
usedNames.add(nome);
const setor=SETORES[Math.floor(r()*SETORES.length)];
const cargo=CARGOS[setor][Math.floor(r()*CARGOS[setor].length)];
const aY=2018+Math.floor(r()*7),aM=1+Math.floor(r()*12),aD=1+Math.floor(r()*28);
const tipoSal=r()>0.8?'Horista':'Mensalista';
const cpf=genCPF(rng(idx*3+1));
const pis=genPIS(rng(idx*5+2));
let situacao='Ativo',termDate=null,feriasStart=null;
if(idx>=5000&&idx<5016){situacao='Desligado';termDate={y:2026,m:5,d:TERM_MAY[idx-5000]}}
else if(idx>=5016&&idx<5030){situacao='Desligado';termDate={y:2026,m:6,d:TERM_JUN[idx-5016]}}
else if(idx>=0&&idx<10){situacao='Férias';feriasStart={y:2026,m:5,d:4}}
else if(idx>=10&&idx<20){situacao='Férias';feriasStart={y:2026,m:6,d:1}}
const pr=r();
let assinaturaStatus;
if(pr<0.644)assinaturaStatus='Assinado';
else if(pr<0.78)assinaturaStatus='Pend. Colaborador';
else if(pr<0.93)assinaturaStatus='Pend. Gestor';
else assinaturaStatus='Pend. RH';
return{idx,matricula:`C${String(idx+1).padStart(5,'0')}`,nome,cpf,pis,cargo,setor,admissao:`${aY}-${fmt2(aM)}-${fmt2(aD)}`,tipoSal,situacao,termDate,feriasStart,assinaturaStatus,seed:Math.floor(r()*999999)};
}

```

■ PONTO

```

function genDayPoint(emp,y,m,d,r){
const dow=new Date(y,m-1,d).getDay();
const we=dow===0||dow===6;
const fKey=`${y}-${fmt2(m)}-${fmt2(d)}`;
const feriado=FERIADOS[fKey];
// desligado após data
if(emp.termDate){const td=emp.termDate;const after=(y>td.y)|| (y===td.y&&m>td.m)|| (y===td.y&&m===td.m&&d>td.d);if(after)return{dow,tipo:'sem-ponto',obs:'Desligado'};if(y===td.y&&m===td.m&&d===td.d)return{dow,tipo:'desligamento',obs:`Desligamento ${fmt2(d)}/${fmt2(m)}`}}
// férias
if(emp.feriasStart){const fs=emp.feriasStart;const st=new Date(fs.y,fs.m-1,fs.d).getTime();const en=st+30*86400000;const dt=new Date(y,m-1,d).getTime();if(dt>=st&&dt<en)return{dow,tipo:'ferias',obs:'Férias'}}
if(we){
if(dow===6&&r()<0.07){const e1=addMin('08:00',rBetween(r,-5,15));const s1=addMin('12:00',rBetween(r,-5,10));const e2=addMin('13:00',rBetween(r,-5,10));const s2=addMin('16:00',rBetween(r,-10,30));const wm=(pTime(s1)-pTime(e1))+(pTime(s2)-pTime(e2));return{dow,tipo:'sabado',e1,s1,e2,s2,workMins:wm,heMins:wm-528,atrasaMins:0,noturno:0,obs:'Sáb. trabalhado'}}
return{dow,tipo:'descanso',obs:dow===0?'Domingo':'Sábado'}}
}
if(feriado){if(r()<0.05){/* trabalhou no feriado */}else return{dow,tipo:'feriado',obs:feriado}}
const rv=r();
if(rv<0.02)return{dow,tipo:'falta',obs:'Falta injustificada'};
if(rv<0.03)return{dow,tipo:'atestado',obs:'Atestado médico'};
if(rv<0.04)return{dow,tipo:'abono',obs:'Abono'};
if(rv<0.055)return{dow,tipo:'esquecimento',e1:addMin('08:00',rBetween(r,-5,10)),obs:'Esq. marcação'};
const e1=addMin('08:00',rBetween(r,-8,15));
const s1=addMin('12:00',rBetween(r,-5,12));
const e2=addMin('13:00',rBetween(r,r()<0.06?15:(-5),r()<0.06?45:15));

```

```

const xr=r();
let s2off=rBetween(r,-10,20);
if(xr<0.15)s2off=rBetween(r,20,60);
else if(xr<0.2)s2off=rBetween(r,-30,-15);
const s2=addMin('17:48',s2off);
const wm=(pTime(s1)-pTime(e1))+(pTime(s2)-pTime(e2));
const he=wm-528;
const noturno=pTime(s2)>pTime('22:00')?pTime(s2)-pTime('22:00'):0;
const atr=pTime(e1)>pTime('08:10')?pTime(e1)-pTime('08:00'):0;
return{dow,tipo:'normal',el,s1,e2,s2,workMins:wm,heMins:he,atrasaMins:atr,noturno,obs:''};
}

function genMonth(emp,y,m){const r=rng(emp.seed+y*100+m);const days=[];for(let
d=1;d<=dimMonth(y,m);d++)days.push(genDayPoint(emp,y,m,d,r));return days}

function computeSummary(data){let he=0,hn=0,fal=0,atr=0,ates=0,abo=0,not=0,fer=0,bankH=0;
for(const d of data){
if(d.tipo==='normal' || d.tipo==='sabado'){if(d.heMins>0)he+=d.heMins;else
if(d.heMins<0)hn+=Math.abs(d.heMins);if(d.atrasaMins>0)atr+=d.atrasaMins;if(d.noturno>0)not+=d.noturno}
if(d.tipo==='falta')fal++;
if(d.tipo==='atestado')ates++;
if(d.tipo==='abono')abo++;
if(d.tipo==='ferias')fer++;
}
}
return{he,hn,fal,atr,ates,abo,not,fer};
}

```

■ STATE

```

let curPage='dashboard',curComp='maio',curCompLabel='Maio/2026',curCompPeriod='01/05/2026 a 31/05/2026';
let tableData=[],filteredData=[],curTblPage=1;
let pendData=[],filteredPend=[],curPendPage=1;
let expData=[],filteredExp=[],curExpPage=1,expSelected=new Set();
const PG=25;
let sortKey='matricula',sortDir=1;
let sumCache={};
function getSummary(emp){
const k=`${emp.idx}-${curComp}`;
if(!sumCache[k]){const[y,m]=curComp==='maio'?[2026,5]:[2026,6];sumCache[k]=computeSummary(genMonth(emp,y,m))}
return sumCache[k];
}

function preload(n=200){for(let i=0;i<Math.min(n,5030);i++){const k=`${i}-${curComp}`;if(!sumCache[k]){const[y,
m]=curComp==='maio'?[2026,5]:[2026,6];sumCache[k]=computeSummary(genMonth(EMPLOYEES[i],y,m))}}}

```

■ NAVIGATION

```

const
PAGE_LABELS={dashboard:'Dashboard',colaboradores:'Colaboradores',pendencias:'Pendências',exportar:'Exportar
Espelhos',jornadas:'Jornadas',parametros:'Parâmetros'};
function navigate(pg){
document.querySelectorAll('.page').forEach(p=>p.classList.remove('active'));
document.querySelectorAll('.sidebar-item').forEach(i=>i.classList.remove('active'));
document.getElementById(`page-${pg}`).classList.add('active');
document.querySelectorAll('.sidebar-item').forEach(i=>{if(i.getAttribute('onclick')==='navigate('${pg}'))i.clas
sList.add('active')});
curPage=pg;
document.getElementById('page-breadcrumb').textContent=PAGE_LABELS[pg]||pg;
if(pg==='colaboradores'){preload(300);buildTableData();filterTable()}
if(pg==='pendencias'){buildPendData();renderPendRows();updatePendKPIs()}
if(pg==='exportar'){preload(300);buildExpData();filterExport()}
}

```

```

function setComp(c){
curComp=c;
curCompLabel=c==='maio'?'Maio/2026':'Junho/2026';
curCompPeriod=c==='maio'?'01/05/2026 a 31/05/2026':'01/06/2026 a 30/06/2026';
document.getElementById('btn-maio').classList.toggle('active',c==='maio');
document.getElementById('btn-junho').classList.toggle('active',c==='junho');
document.getElementById('dash-comp-label').textContent=curCompLabel;
document.getElementById('colab-comp-label').textContent=curCompLabel;
updateDashCards();renderCharts();
if(curPage==='colaboradores'){buildTableData();filterTable()}
if(curPage==='pendencias'){buildPendData();renderPendRows();updatePendKPIs()}
if(curPage==='exportar'){buildExpData();filterExport()}
}

```

■ DASHBOARD

```

function updateDashCards(){
const m=curComp==='maio';
document.getElementById('card-desl').textContent=m?'16':'14';
document.getElementById('card-ferias').textContent='10';
document.getElementById('card-he').textContent=m?'1.847h':'2.103h';
document.getElementById('card-hn').textContent=m?'389h':'412h';
document.getElementById('card-ass').textContent=m?'3.241':'2.988';
document.getElementById('card-pend').textContent=m?'1.789':'2.042';
}

function updatePendKPIs(){
const m=curComp==='maio';
document.getElementById('pend-count').textContent=m?'1.789':'2.042';
document.getElementById('pend-colab').textContent=m?'1.102':'1.287';
document.getElementById('pend-gestor').textContent=m?'687':'755';
document.getElementById('pend-rh').textContent=m?'347':'412';
}

let charts={};
function dc(id){if(charts[id]){charts[id].destroy();delete charts[id]}}
function renderCharts(){
const m=curComp==='maio';
const slabels=SETORES.slice(0,10);
const she=m?[312,87,143,201,78,167,95,124,45,89]:[341,92,167,218,83,178,102,139,51,94];
dc('chartSetores');charts['chartSetores']=new Chart(document.getElementById('chartSetores'),{type:'bar',data:{labels:slabels,datasets:[{label:'H.Extras',data:she,backgroundColor:'rgba(26,86,219,.8)',borderRadius:4}],options:{plugins:{legend:{display:false}},responsive:true,maintainAspectRatio:false,scales:{y:{grid:{color:'#f1f5f9'}},ticks:{font:{size:10}}},x:{ticks:{font:{size:9},maxRotation:30}}}}});
dc('chartOcorrencias');charts['chartOcorrencias']=new Chart(document.getElementById('chartOcorrencias'),{type:'doughnut',data:{labels:['H.Extras','Faltas','Atestados','Abonos','Atrasos','Esq.Marc.'],datasets:[{data:m?[1847,312,189,98,445,234]:[2103,289,201,112,398,198],backgroundColor:['#1a56db','#ef4444','#f59e0b','#10b981','#8b5cf6','#64748b']}],options:{plugins:{legend:{position:'bottom',labels:{font:{size:9},padding:8}},responsive:true,maintainAspectRatio:false}}});
const dias=Array.from({length:dimMonth(2026,m?5:6)},(_ ,i)=>i+1).filter(d=>!isWE(2026,m?5:6,d));
const r2=rng(m?1:2);
dc('chartDiario');charts['chartDiario']=new Chart(document.getElementById('chartDiario'),{type:'line',data:{labels:dias.map(String),datasets:[{label:'H.Extras',data:dias.map(()=>Math.floor(r2()*80+20)),borderColor:'#1a56db',backgroundColor:'rgba(26,86,219,.1)',borderWidth:2,pointRadius:2,fill:true,tension:.3}],options:{plugins:{legend:{display:false}},responsive:true,maintainAspectRatio:false,scales:{y:{grid:{color:'#f1f5f9'}},ticks:{font:{size:10}}},x:{ticks:{font:{size:9}}}}}}});
const sem=m?['01-09/Mai','10-16/Mai','17-23/Mai','24-31/Mai']:['01-06/Jun','07-13/Jun','14-20/Jun','21-30/Jun'];
dc('chartFaltas');charts['chartFaltas']=new Chart(document.getElementById('chartFaltas'),{type:'bar',data:{labels:sem,datasets:[{label:'Faltas',data:m?[78,92,67,75]:[81,88,71,69],backgroundColor:'rgba(239,68,68,.8)',border

```

```

Radius:4]]},options:{plugins:{legend:{display:false}},responsive:true,maintainAspectRatio:false,scales:{y:{grid
:{color:'#f1f5f9'},ticks:{font:{size:10}}},x:{ticks:{font:{size:9}}}}}});

dc('chartCargos');charts['chartCargos']=new Chart(document.getElementById('chartCargos'),{type:'pie',data:{labe
ls:SETORES.slice(0,8).map(s=>s.length>14?s.slice(0,14)+'...':s),datasets:[{data:[820,312,445,680,234,512,389,278]
,backgroundColor:['#1a56db','#10b981','#f59e0b','#ef4444','#8b5cf6','#0891b2','#d97706','#6366f1']}}},options:{
plugins:{legend:{position:'bottom',labels:{font:{size:8,padding:5}}},responsive:true,maintainAspectRatio:false
}});

const top10=EMPLOYEES.slice(50,60).map((e,i)=>({n:e.nome.split(' ')[0]+' '+e.nome.split('
')[2],v:80-i*5+Math.floor(rng(i+99)()*8)}));

dc('chartTop10');charts['chartTop10']=new Chart(document.getElementById('chartTop10'),{type:'bar',data:{labels:
top10.map(x=>x.n),datasets:[{label:'H.Extras',data:top10.map(x=>x.v),backgroundColor:'rgba(16,185,129,.8)',bord
erRadius:4}}},options:{indexAxis:'y',plugins:{legend:{display:false}},responsive:true,maintainAspectRatio:false
,scales:{x:{grid:{color:'#f1f5f9'},ticks:{font:{size:10}}},y:{ticks:{font:{size:9}}}}}});

dc('chartPendencias');charts['chartPendencias']=new
Chart(document.getElementById('chartPendencias'),{type:'doughnut',data:{labels:['Assinado','Pend.
Collab','Pend. Gestor','Pend. RH'],datasets:[{data:m?[3241,1102,687,347]:[2988,1287,755,412],backgroundColor:['
#10b981','#f59e0b','#3b82f6','#ef4444']}}},options:{plugins:{legend:{position:'bottom',labels:{font:{size:9},pa
dding:8}}},responsive:true,maintainAspectRatio:false}});

}

```

■ TABLE COLABORADORES

```

function buildTableData(){
tableData=EMPLOYEES.map(e=>{const s=getSummary(e);return {...e,he:s.he,hn:s.hn}});
const setores=[...new Set(EMPLOYEES.map(e=>e.setor))].sort();
const cargos=[...new Set(EMPLOYEES.map(e=>e.cargo))].sort();
document.getElementById('filterSetor').innerHTML='<option value="">Todos os
setores</option>'+setores.map(s=>'<option>${s}</option>').join('');
document.getElementById('filterCargo').innerHTML='<option value="">Todos os
cargos</option>'+cargos.map(c=>'<option>${c}</option>').join('');
}

function filterTable(){
const q=document.getElementById('searchInput').value.toLowerCase();
const st=document.getElementById('filterSetor').value;
const cg=document.getElementById('filterCargo').value;
const si=document.getElementById('filterSit').value;
const pe=document.getElementById('filterPend').value;
filteredData=tableData.filter(e=>{
if(q&&!e.nome.toLowerCase().includes(q)&&!e.matricula.toLowerCase().includes(q))return false;
if(st&&e.setor!==st)return false;
if(cg&&e.cargo!==cg)return false;
if(si&&e.situacao!==si)return false;
if(pe==='Pendente'&&e.assinaturaStatus==='Assinado')return false;
if(pe==='Assinado'&&e.assinaturaStatus!=='Assinado')return false;
return true;
});
curTblPage=1;renderTblRows();
}

function sortTable(k){if(sortKey===k)sortDir*=-1;else{sortKey=k;sortDir=1}filteredData.sort((a,b)=>typeof
a[k]=== 'string'?a[k].localeCompare(b[k])*sortDir:(a[k]-b[k])*sortDir);renderTblRows()}

function renderTblRows(){
const st=(curTblPage-1)*PG,en=st+PG,rows=filteredData.slice(st,en);
const BADGES={Ativo:'badge-green',Desligado:'badge-red','Férias':'badge-yellow'};
const PBADGES={Assinado:'badge-green','Pend. Colaborador':'badge-yellow','Pend. Gestor':'badge-blue','Pend.
RH':'badge-red'};
document.getElementById('tbody').innerHTML=rows.map(e=>'<tr>
<td><span class="mat-code">${e.matricula}</span></td>
<td class="fw-600">${e.nome}</td>
<td>${e.cargo}</td><td>${e.setor}</td>

```

```

<td><span class="badge ${BADGES[e.situacao]||'badge-gray'}"><span
class="badge-dot"></span>${e.situacao}</span></td>
<td class="text-right ${e.he>0?'color-success':''}">${e.he>0?''+fmtH(e.he):-'}</td>
<td class="text-right ${e.hn>0?'color-danger':''}">${e.hn>0?'-'+fmtH(e.hn):-'}</td>
<td><span class="badge ${PBADGES[e.assinaturaStatus]||'badge-gray'}">${e.assinaturaStatus}</span></td>
<td><button class="btn btn-primary" style="padding:4px 10px;font-size:11px" onclick="openMirror(${e.idx})"><svg
width="12" height="12" fill="none" stroke="currentColor" viewBox="0 0 24 24" stroke-width="2"><path d="M1 12s4-8
11-8 11 8 11 8-4 8-11 8-11-8-11-8z"/></svg>Ver</button></td>
</tr>`).join('');
const tot=filteredData.length;
document.getElementById('pag-info').textContent=`Mostrando ${st+1}-${Math.min(en,tot)} de
${tot.toLocaleString('pt-BR')}`;
renderPagBtns('pag-btns',curTblPage,Math.ceil(tot/PG),'goTbl');
}
function goTbl(p){const
max=Math.ceil(filteredData.length/PG);if(p<1||p>max)return;curTblPage=p;renderTblRows()}

```

■ TABLE PENDÊNCIAS

```

function
buildPendData(){pendData=EMPLOYEES.filter(e=>e.assinaturaStatus!=='Assinado');filteredPend=[...pendData]}
function filterPendTable(){const q=document.getElementById('searchPend').value.toLowerCase();filteredPend=pendD
ata.filter(e=>!q||e.nome.toLowerCase().includes(q)||e.matricula.toLowerCase().includes(q));curPendPage=1;render
PendRows()}
function renderPendRows(){
const st=(curPendPage-1)*PG,en=st+PG,rows=filteredPend.slice(st,en);
document.getElementById('tbody-pend').innerHTML=rows.map(e=>{
const sc=e.assinaturaStatus.includes('Colaborador');const sg=e.assinaturaStatus.includes('Gestor');const
sr=e.assinaturaStatus.includes('RH');
return`<tr><td><span class="mat-code">${e.matricula}</span></td><td
class="fw-600">${e.nome}</td><td>${e.setor}</td><td>${e.cargo}</td>
<td><span class="badge ${sc?'badge-yellow':'badge-green'}">${sc?'Pendente':'Assinado'}</span></td>
<td><span class="badge
${(!sc&&sg)?'badge-blue':'badge-gray'}">${(!sc&&sg)?'Pendente':sc?'Aguardando':'Aprovado'}</span></td>
<td><span class="badge
${(!sc&&!sg&&sr)?'badge-red':'badge-gray'}">${(!sc&&!sg&&sr)?'Pendente':'Aguardando'}</span></td>
<td><button class="btn btn-secondary" style="padding:4px 10px;font-size:11px"
onclick="openMirror(${e.idx})">Ver</button></td></tr>`;
}).join('');
const tot=filteredPend.length;
document.getElementById('pag-pend-info').textContent=`Mostrando ${ (curPendPage-1)*PG+1}-${Math.min(en,tot)} de
${tot.toLocaleString('pt-BR')}`;
renderPagBtns('pag-pend-btns',curPendPage,Math.ceil(tot/PG),'goPend');
}
function goPend(p){const
max=Math.ceil(filteredPend.length/PG);if(p<1||p>max)return;curPendPage=p;renderPendRows()}

```

■ EXPORTAR

```

function buildExpData(){
expData=EMPLOYEES.map(e=>{const s=getSummary(e);return{...e,he:s.he,hn:s.hn}});
const setores=[...new Set(EMPLOYEES.map(e=>e.setor))].sort();
document.getElementById('expSetor').innerHTML='<option value="">Todos os
setores</option>'+setores.map(s=>`<option>${s}</option>`).join('');
expSelected=new Set();updateExpSel();
}
function filterExport(){
const q=document.getElementById('expSearch').value.toLowerCase();
const st=document.getElementById('expSetor').value;
const si=document.getElementById('expSit').value;

```

```

filteredExp=expData.filter(e=>{
if(q&&!e.nome.toLowerCase().includes(q)&&!e.matricula.toLowerCase().includes(q))return false;
if(st&&e.setor!=st)return false;
if(si&&e.situacao!=si)return false;
return true;
});
// remove sel fora do filtro
const ids=new Set(filteredExp.map(e=>e.idx));
for(const id of expSelected)if(!ids.has(id))expSelected.delete(id);
curExpPage=1;renderExpRows();updateExpSel();
}
function updateExpSel(){
const n=expSelected.size;
document.getElementById('exp-found').textContent=filteredExp.length.toLocaleString('pt-BR');
document.getElementById('exp-sel').textContent=n;
document.getElementById('exp-pdf-n').textContent=n;
document.getElementById('btn-exp-pdf').disabled=n===0;
document.getElementById('btn-exp-csv').disabled=n===0;
const chkAll=document.getElementById('chk-exp-all');
if(chkAll)chkAll.checked=filteredExp.length>0&&filteredExp.every(e=>expSelected.has(e.idx));
}
function renderExpRows(){
const st=(curExpPage-1)*PG,en=st+PG,rows=filteredExp.slice(st,en);
const BADGES={Ativo:'badge-green',Desligado:'badge-red','Férias':'badge-yellow'};
const PBADGES={Assinado:'badge-green','Pend. Colaborador':'badge-yellow','Pend. Gestor':'badge-blue','Pend. RH':'badge-red'};
document.getElementById('tbody-exp').innerHTML=rows.map(e=>`<tr>
<td><input type="checkbox" ${expSelected.has(e.idx)?'checked':''}>
onchange="expToggleOne(${e.idx},this.checked)"></td>
<td><span class="mat-code">${e.matricula}</span></td>
<td class="fw-600">${e.nome}</td>
<td>${e.cargo}</td><td>${e.setor}</td>
<td><span class="badge ${BADGES[e.situacao]}|'|'badge-gray'"><span
class="badge-dot"></span>${e.situacao}</span></td>
<td>${curCompLabel}</td>
<td><span class="badge ${PBADGES[e.assinaturaStatus]}|'|'badge-gray'">${e.assinaturaStatus}</span></td>
<td><button class="btn btn-primary" style="padding:4px 10px;font-size:11px" onclick="openMirror(${e.idx})"><svg
width="12" height="12" fill="none" stroke="currentColor" viewBox="0 0 24 24" stroke-width="2"><path d="M1 12s4-8
11-8 11 8-4 8-11 8-11-8z"/><circle cx="12" cy="12" r="3"/></svg>Ver</button>
<button class="btn btn-secondary" style="padding:4px 8px;font-size:11px;margin-left:4px"
onclick="exportOnePDF(${e.idx})"><svg width="12" height="12" fill="none" stroke="currentColor" viewBox="0 0 24
24" stroke-width="2"><path d="M14 2H6a2 2 0 0 0-2 2v16a2 2 0 0 0 2 2h12a2 2 0 0 0 2-2V8z"/><polyline points="14
2 14 8 20 8"/></svg>PDF</button></td>
</tr>`).join('');
const tot=filteredExp.length;
document.getElementById('pag-exp-info').textContent='Mostrando ${st+1}-${Math.min(en,tot)} de
${tot.toLocaleString('pt-BR')}`;
renderPagBtns('pag-exp-btns',curExpPage,Math.ceil(tot/PG),'goExp');
}

```

■ PAGINATION HELPER

```

function renderPagBtns(containerId,cur,total,fn){
let h=`<button class="page-btn" onclick="${fn}(${cur-1})" ${cur<=1?'disabled':''}></button>`;
const ar=2;
for(let p=1;p<=total;p++){
if(p===1||p===total||Math.abs(p-cur)<=ar)h+=`<button class="page-btn ${p===cur?'active':''}"
onclick="${fn}(${p})">${p}</button>`;
}

```

```
else if(Math.abs(p-cur)==ar+1)h+=`<button class="page-btn" disabled>...</button>`;
}
h+=`<button class="page-btn" onclick="$ {fn} ($ {cur+1})" $ {cur}>total?'disabled':''>></button>`;
document.getElementById(containerId).innerHTML=h;
}
```

■ ESPELHO — HTML FIEL AO MODELO

[illegible]


```

return`<tr class="${cls}"><td>${dt}</td><td>${dn}</td><td>08:00 17:48</td><td>60</td><td>${d.e1}</td><td>${d.s1}</td><td>${d.e2}</td><td>${d.s2}</td><td>${wStr}</td><td>${heStr}</td><td>${hnStr}</td><td>${ntStr}</td><td class="oc">${ocStr}</td><td></td><td></td><td></td></tr>`;
}).join('');

// Total periodo
const workDays=data.filter(d=>d.tipo==='normal' || d.tipo==='sabado').length;
const totalH=fmtH(workDays*528+sum.he-sum.hn);

const he50=Math.floor(sum.he*0.6),he100=sum.he-he50;

// Resumo pagamentos (fiel ao modelo)
const resumoRows=[
sum.fer>0?`<div class="ep-resumo-row"><div>00006</div><div>FÉRIAS</div><div>DIAS</div><div>${sum.fer.toFixed(2)}</div><div>${fmtH(sum.fer*528)}</div></div>`,`',
sum.ates>0?`<div class="ep-resumo-row"><div>00019</div><div>ATESTADOS</div><div>${(sum.ates*8.8).toFixed(2)}</div><div>${fmtH(sum.ates*528)}</div></div>`,`',
sum.fal>0?`<div class="ep-resumo-row"><div>00022</div><div>DESC. FALTAS E ATRASOS INJUSTIFICADOS</div><div>${(sum.fal*8.8+sum.atr/60).toFixed(2)}</div><div>${fmtH(sum.fal*528+sum.atr)}</div></div>`,`',
`<div class="ep-resumo-row"><div>00086</div><div>EXTRAS DOM E FER</div><div>5.50</div><div>05:30</div></div>`,`',
sum.not>0?`<div class="ep-resumo-row"><div>00092</div><div>EXTRAS 100% NOTURNAS</div><div>${(sum.not/60).toFixed(2)}</div><div>${fmtH(sum.not)}</div></div>`,`',
he100>0?`<div class="ep-resumo-row"><div>00096</div><div>EXTRAS 100%</div><div>${(he100/60).toFixed(2)}</div><div>${fmtH(he100)}</div></div>`,`',
he50>0?`<div class="ep-resumo-row"><div>00099</div><div>EXTRAS 50%</div><div>${(he50/60).toFixed(2)}</div><div>${fmtH(he50)}</div></div>`,`',
].filter(Boolean).join('');

return`<div class="ep-wrap">

```

■ OPEN MIRROR MODAL

```

function openMirror(empIdx){
const emp=EMPLOYEES[empIdx];
document.getElementById('modal-title').textContent=`Espelho de Ponto - ${emp.nome} - ${curCompLabel}`;
document.getElementById('modal-body').innerHTML=`<div style="background:#fff;padding:0">${buildMirrorHTML(emp)}</div>`;
document.getElementById('mirror-overlay').classList.add('open');

// store for print
document.getElementById('mirror-overlay').dataset.idx=empIdx;
}

function closeMirror(){document.getElementById('mirror-overlay').classList.remove('open')}
function printMirror(){
const idx=parseInt(document.getElementById('mirror-overlay').dataset.idx);
exportOnePDF(idx);
}

```

■ PRINT WINDOW

```

const PRINT_CSS=`
body{font-family:Arial,Helvetica,sans-serif;font-size:9px;margin:8px;background:#fff}
.ep-wrap{background:#fff;font-family:Arial,Helvetica,sans-serif;font-size:9px;width:100%;border:1px solid #888;padding:6px}
.ep-header{border:1px solid #999;margin-bottom:2px}
.ep-logo-row{display:flex;align-items:stretch;border-bottom:1px solid #999}
.ep-logo-box{background:#1a3f7a;display:flex;align-items:center;justify-content:center;padding:6px 10px;min-width:90px}
.ep-logo-text{color:#fff;font-size:11px;font-weight:800;line-height:1.1;text-align:center}
.ep-logo-sub{color:#a0c4ff;font-size:7px;font-weight:400}
.ep-title-box{flex:1;display:flex;flex-direction:column;align-items:center;justify-content:center;padding:4px}
.ep-title{font-size:13px;font-weight:700;color:#000}
.ep-comp-line{font-size:9px;color:#333;margin-top:2px}
.ep-page-box{text-align:right;padding:4px 8px;font-size:9px;color:#333;white-space:nowrap}
.ep-empresa-row{padding:4px 8px;font-size:9px;font-weight:700;color:#000;border-bottom:1px solid #ccc}

```

```

.ep-cnpj-row{padding:3px
8px;font-size:8px;color:#333;display:flex;justify-content:space-between;border-bottom:1px solid #ccc}
.ep-fl{font-size:7px;font-weight:700;color:#333;text-transform:uppercase;padding:2px 6px;border-right:1px solid
#ccc;border-bottom:1px solid #ccc;white-space:nowrap;background:#f5f5f5}
.ep-fv{font-size:9px;font-weight:600;color:#000;padding:2px 6px;border-right:1px solid #ccc;border-bottom:1px
solid #ccc}
.ep-name-row{display:flex;border-bottom:1px solid #ccc}
.ep-name-l{font-size:7px;font-weight:700;color:#333;text-transform:uppercase;padding:2px 6px;border-right:1px
solid #ccc;background:#f5f5f5;white-space:nowrap}
.ep-name-v{font-size:9px;font-weight:700;color:#000;padding:2px 6px;flex:1;border-right:1px solid #ccc}
.ep-pis-l{font-size:8px;color:#333;padding:2px 8px;white-space:nowrap}
.ep-table{width:100%;border-collapse:collapse;margin-top:3px;font-size:7.5px}
.ep-table th{background:#1a3f7a;color:#fff;font-size:7px;font-weight:700;padding:3px
3px;text-align:center;border:1px solid #0f2a5a}
.ep-table th.th-jornada{background:#2a5298}
.ep-table td{font-size:7.5px;padding:2px 3px;text-align:center;border:1px solid #ccc;color:#000}
.ep-table tr.we td{background:#eeeeee;color:#888}
.ep-table tr.fer td{background:#fff7e0}
.ep-table tr.ferias-row td{background:#e8f5e9}
.ep-table tr.falta-row td{background:#fff0f0}
.ep-table tr.even td{background:#f7f9fc}
.ep-table td.oc{text-align:left;font-size:7px}
.ep-resumo{margin-top:4px;border:1px solid #999}
.ep-resumo-title{background:#1a3f7a;color:#fff;font-size:8px;font-weight:700;padding:3px
6px;display:grid;grid-template-columns:60px 1fr 80px 80px}
.ep-resumo-row{display:grid;grid-template-columns:60px 1fr 80px 80px;border-top:1px solid #ddd;font-size:8px}
.ep-resumo-row div{padding:2px 6px;border-right:1px solid #ddd}
.ep-sign{margin-top:20px;display:flex;justify-content:center}
.ep-sign-line{text-align:center}
.ep-sign-rule{border-top:1px solid #000;width:280px;margin:0 auto 4px}
.ep-sign-label{font-size:8px;color:#333}
@media print{body{margin:0}@page{margin:10mm}}
function openPrintWindow(html,title){
const w=window.open('','_blank','width=950,height=750');
w.document.write(`<!DOCTYPE html><html><head><meta
charset="UTF-8"><title>${title}</title><style>${PRINT_CSS}</style></head><body>${html}</body></html>`);
w.document.close();
setTimeout(()=>w.print(),600);
}

```

■ CLOSE MODAL ON OVERLAY CLICK

```
document.getElementById('mirror-overlay').addEventListener('click',function(e){if(e.target===this)closeMirror()
});
```

■ INIT

```
document.addEventListener('DOMContentLoaded',()=>{updateDashCards();renderCharts()});
```